



Linear Systems Theory

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Module 3

Lecture 3

Math Preliminaries: Linear Algebra



Diagonalisation

Diagonal Matrices and Properties

A square matrix with non-diagonal entries being zeros

$$a_{ij} = 0 \quad \forall i \neq j$$

Properties of diagonal matrices:

- ▶ Rank of a diagonal matrix is equal to its non-zero diagonal entries
- ▶ The diagonal elements of a diagonal matrix are its Eigen values
- ▶ Product of two diagonal matrices **A** and **B** is another diagonal matrix and is also commutative : **AB = BA**
- ▶ The n^{th} power of a diagonal matrix is also diagonal and $[\mathbf{A}^n]_{ii} = a_{ii}^n$
- ▶ The inverse of a diagonal matrix with non-zero diagonal entries is also diagonal and its diagonal elements are:

$$[\mathbf{A}^{-1}]_{ii} = \frac{1}{a_{ii}}$$



Matrix Diagonalization

If a square matrix $\mathbf{A}$ is similar to a diagonal matrix $\mathbf{D}$, then it is said to be diagonalizable

- ▶ Alternately, for a linear transformation $f: V \rightarrow V$ represented by $\mathbf{A}$, there exists a basis for V which results in a diagonal matrix $\mathbf{D}$ corresponding to f
- ▶ That basis for V is given by the eigen vectors of $\mathbf{A}$
- ▶ Let $\lambda_1, \dots, \lambda_n$ be the eigen values and $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the eigen vectors of an $n \times n$ square matrix $\mathbf{A}$ which is diagonalizable
- ▶ Then $\mathbf{e}_1, \dots, \mathbf{e}_n$ form a basis for $\mathbb{R}^n$
- ▶ Let $\mathbf{D} = \text{diag} \begin{bmatrix} \lambda_1 & \dots & \lambda_n \end{bmatrix}$ and $\mathbf{E}$ be a matrix with eigen vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ as its columns, then

$$\mathbf{A}\mathbf{E} = \mathbf{E}\mathbf{D} \implies \mathbf{D} = \mathbf{E}^{-1}\mathbf{A}\mathbf{E}$$

- ▶ **Theorem:** A $n \times n$ square matrix is diagonalizable if and only if it has n linearly independent eigen vectors



Matrix Diagonalization: Example

Eg. Diagonalise the matrix $\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 0 & -1 \end{bmatrix}$

Solution: $\lambda = -1, 2$ and the eigen vectors are $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Exercise 1

Diagonalise the matrix $\mathbf{A} = \begin{bmatrix} 6 & 2 \\ -1 & 3 \end{bmatrix}$



Defective Matrices and Generalised Eigen Vectors

Defective Matrices

Matrices with eigen values whose algebraic multiplicity exceeds the geometric multiplicity are **defective**.

- ▶ Eg. $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$; $\lambda = 3, 3$ For $\lambda = 3$, algebraic multiplicity is 2 and geometric multiplicity is 1
- ▶ Defective matrices of size $n \times n$ have less than n linearly independent eigen vectors
- ▶ Therefore defective matrices are not diagonalizable
- ▶ Then, how to get a basis for $\mathbb{R}^n$ comprising of eigen vectors of $\mathbf{A}$?
 - **Generalised eigen vectors**



Generalised Eigen Vector

A generalised eigen vector of an $n \times n$ matrix $\mathbf{A}$ corresponding to the eigen value λ is a non-zero vector $\mathbf{x}$ satisfying:

$$(\mathbf{A} - \lambda \mathbf{I}_n)^p \mathbf{x} = \mathbf{0} \quad (1)$$

where p is a positive integer. Equivalently, it is a non-zero element of the nullspace of $(\mathbf{A} - \lambda \mathbf{I}_n)^p$.

- ▶ The value of p corresponding to an eigen value λ can be atmost the algebraic multiplicity of λ denoted by k_λ
- ▶ Eigen vectors are generalised eigen vectors with $p = 1$
- ▶ Generalised eigen vectors form an eigen vector basis for $\mathbb{R}^n$ in case the matrix is defective



Finding Generalised Eigen Vectors

For an $n \times n$ matrix, let the algebraic multiplicity of an eigen value λ be k_λ .

$$(\mathbf{A} - \lambda \mathbf{I}_n) \mathbf{x}_p = \mathbf{x}_{p-1} \quad \forall p = 2, \dots, k_\lambda \quad (2)$$

- ▶ Let $\mathbf{x}_1$ be the eigen vector corresponding to $p = 1$, that is, $(\mathbf{A} - \lambda \mathbf{I}_n) \mathbf{x}_1 = \mathbf{0}$
- ▶ As per Eq.2,

$$\begin{aligned} (\mathbf{A} - \lambda \mathbf{I}_n) \mathbf{x}_2 &= \mathbf{x}_1 \\ (\mathbf{A} - \lambda \mathbf{I}_n)^2 \mathbf{x}_2 &= (\mathbf{A} - \lambda \mathbf{I}_n) \mathbf{x}_1 = \mathbf{0} \end{aligned}$$

Therefore, solving Eq.2 is equivalent to finding $\mathbf{x}$ in Eq.1, for different values of p

- ▶ In this manner, generalised eigen vectors form a chain:

$$\mathbf{x}_p \xrightarrow{\mathbf{A} - \lambda \mathbf{I}} \mathbf{x}_{p-1} \xrightarrow{\mathbf{A} - \lambda \mathbf{I}} \dots \xrightarrow{\mathbf{A} - \lambda \mathbf{I}} \mathbf{x}_2 \xrightarrow{\mathbf{A} - \lambda \mathbf{I}} \mathbf{x}_1 \xrightarrow{\mathbf{A} - \lambda \mathbf{I}} \mathbf{0}$$



Generalised Eigen Vector: Example

► Eg. $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find a basis for $\mathbb{R}^2$ comprising of generalised eigen vectors of $\mathbf{A}$.

Solution: $\lambda = 1, 1$



Block Matrices and Jordan Normal Forms

Block Matrix

A block matrix is a matrix defined using smaller matrices or submatrices called blocks.
Egs:

$$X_{4 \times 4} = \begin{bmatrix} A_{2 \times 2} & B_{2 \times 2} \\ C_{2 \times 2} & D_{2 \times 2} \end{bmatrix}; X_{4 \times 4} = \begin{bmatrix} A_{1 \times 1} & B_{1 \times 3} \\ C_{3 \times 1} & D_{3 \times 3} \end{bmatrix}$$

where **A, B, C, D** are submatrices and referred to as blocks of **X**

- ▶ The submatrices can be of any dimension as long as they are together in consistence with the block matrix
- ▶ The blocks along the diagonal are referred to as diagonal blocks
- ▶ Two block matrices of size and square diagonal blocks multiply similar to matrix multiplication

$$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} A_1 A_2 + B_1 C_2 & A_1 B_2 + B_1 D_2 \\ C_1 A_2 + D_1 C_2 & C_1 B_2 + D_1 D_2 \end{bmatrix}$$



Block Diagonal Matrix

A block matrix which is a square matrix, and whose diagonal blocks are square matrices and off-diagonal blocks are zero matrices. Eg:

$$A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$$

Properties of block diagonal matrices with n diagonal blocks:

- ▶ $\det(A) = \det(A_1) \det(A_2) \dots \det(A_n)$
- ▶ $\text{tr}(A) = \text{tr}(A_1) + \text{tr}(A_2) + \dots + \text{tr}(A_n)$
- ▶ The inverse of a block diagonal matrix is another block diagonal matrix, composed of the inverse of each diagonal block: $A^{-1} = \text{diag}(A_1^{-1}, \dots, A_n^{-1})$
- ▶ The eigen values of a block diagonal matrix A are combination of eigen values of $A_1, \dots, A_n$



Jordan Normal Form or Jordan Canonical Form or Jordan Matrix

A Jordan matrix is a block diagonal matrix whose diagonal blocks are all Jordan blocks. It is a generalisation of diagonal matrices

- ▶ A Jordan block associated with λ is a square, upper triangular matrix whose entries are all λ on the diagonal, all 1 on the entries immediately above the diagonal, and 0 elsewhere.

$$\text{Eg. } J = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

- ▶ In a Jordan matrix, each jordan block can be associated with a different value of λ
- ▶ Eg. of a Jordan Matrix with two Jordan blocks of size 3×3 and 1×1 corresponding to $\lambda = 2$ and $\lambda = 3$

$$J = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$



Properties of Jordan Matrices

- ▶ Diagonal matrix is a special case of Jordan matrix with each diagonal element forming a Jordan block
- ▶ There exists a Jordan matrix that is similar to every square matrix i.e., every square matrix $\mathbf{A}$ can be transformed into a Jordan matrix.

$$\mathbf{J} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

- ▶ The λ values of Jordan blocks in $\mathbf{J}$ are given by the eigen values of $\mathbf{A}$
- ▶ Modal matrix $\mathbf{P}$ comprises of the generalised eigen vectors of $\mathbf{A}$ as its columns



Jordan Matrices: Example

Find the Jordan form of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

Solution: $\lambda = 3, 3$



Symmetric Matrices

Symmetric Matrices and Properties

Symmetric matrix is a square matrix that is equal to its transpose

$$\mathbf{A} = \mathbf{A}^T$$

$$a_{ij} = a_{ji}$$

Properties of symmetric matrices:

- ▶ Every square diagonal matrix is symmetric
- ▶ The sum and difference of two symmetric matrices is symmetric
- ▶ $\mathbf{A}^n$ is symmetric, if $\mathbf{A}$ is symmetric
- ▶ If $\mathbf{A}^{-1}$ exists, it is symmetric iff $\mathbf{A}$ is symmetric
- ▶ The eigen values of a symmetric matrix are all real
- ▶ The eigen vectors corresponding to distinct eigen values of a symmetric matrix are orthogonal to each other
- ▶ Every symmetric matrix is diagonalizable i.e., its eigen vectors are linearly independent



Eigen Decomposition and Singular Value Decomposition (SVD)

Eigen Decomposition

A decomposition which decomposes a square matrix $\mathbf{A}$ into a set of its eigen values and eigen vectors

$$\mathbf{A} = \mathbf{E}\mathbf{D}\mathbf{E}^{-1}$$

where $\mathbf{E}$ is a matrix with eigen vectors of $\mathbf{A}$ as its columns and $\mathbf{D}$ is a diagonal matrix with eigen values of $\mathbf{A}$ as diagonal elements

- ▶ It is an alternate interpretation to matrix diagonalization
- ▶ Eigen decomposition is possible only if the eigen vectors of $\mathbf{A}$ are linearly independent
- ▶ This decomposition has its applications in calculating powers of a matrix and matrix exponential

$$\mathbf{A}^n = \mathbf{E}\mathbf{D}^n\mathbf{E}^{-1}$$

$$e^{\mathbf{A}} = \mathbf{E}e^{\mathbf{D}}\mathbf{E}^{-1}$$

- ▶ What if $\mathbf{A}$ is rectangular ? - **Singular Value Decomposition**



Singular Value Decomposition (SVD)

Singular Values of a Matrix

- ▶ Let $A \in \mathbb{R}^{m \times n}$; $\text{rank}(A) \leq \min(m, n)$
- ▶ $AA^T \in \mathbb{R}^{m \times m}$, is a symmetric matrix.
- ▶ Let $\lambda_1, \dots, \lambda_r$, $r \leq m$ denote the non zero (these are all real) eigen values of AA^T .
 $\text{rank}(A) = r$
- ▶ The non zero singular values of A are $\sqrt{\lambda_i}$, $i = 1, \dots, r$.
- ▶ The remaining singular values of A are zero.



Singular Value Decomposition (SVD)

The Singular Value decomposition is the decomposition of a matrix into three other matrices:

$$\mathbf{A}_{m \times n} = \mathbf{U}_{m \times m} \mathbf{S}_{m \times n} \mathbf{V}_{n \times n}^T$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices and $\mathbf{S}$ is a matrix containing elements along the main diagonal

- ▶ This is a generalisation of eigen decomposition to any $m \times n$ matrix
- ▶ The diagonal elements of $\mathbf{S}$ are the singular values of $\mathbf{A}$
- ▶ The columns of $\mathbf{U}$ are the eigen vectors of $\mathbf{A}\mathbf{A}^T$ (or) the left singular vectors of $\mathbf{A}$
- ▶ The columns of $\mathbf{V}$ are the eigen vectors of $\mathbf{A}^T\mathbf{A}$ (or) the right singular vectors of $\mathbf{A}$



Singular Value Decomposition (SVD)

- ▶ SVD and Eigen decomposition coincide if and only if $\mathbf{A}$ is symmetric and positive definite
- ▶ An orthogonal basis for each of the fundamental subspaces of a matrix $\mathbf{A}$ is given by Singular Value Decomposition (SVD)
 - Basis for $\mathcal{C}(\mathbf{A})$: First r columns of $\mathbf{U}$
 - Basis for $\mathcal{N}(\mathbf{A})$: Last $n - r$ columns of $\mathbf{V}$
 - Basis for $\mathcal{R}(\mathbf{A})$: First r columns of $\mathbf{V}$
 - Basis for $\mathcal{N}(\mathbf{A}^T)$: Last $m - r$ columns of $\mathbf{U}$



Summary: Mod 3 Lecture 3

- ▶ Diagonalisation
- ▶ Generalised Eigen vectors
- ▶ Jordan Matrices
- ▶ Singular Value Decomposition

Contents: Module 4

- ▶ State space models
- ▶ State transition matrix
- ▶ Solutions to LTI and LTV systems

